

Toward an Independent AthenaK Reproduction of CGL–Landau-Fluid Self-Organization in High- β Turbulence

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Abstract

Pressure anisotropy may reorganize collisionless, high- β turbulence before microscale instabilities provide an effective collisional closure, but the numerical evidence for this “magneto-immutable” response is demanding to reproduce. We define an independent AthenaK reproduction of the CGL–Landau-fluid (CGL–LF) simulations of Majeski, Kunz, and Squire (2024, MKS24), after validating the heat-flux operator, threshold policy, driven turbulence problem generator, and retained diagnostics. The reproduction targets sixteen unique source-mapped CGL–LF calculations spanning active/passive pressure feedback, forcing geometry, beta, heat-flux scale, limiter scattering, and scale separation. Several dimensionless published products admit checksum-qualified digitized comparisons, whereas absolute spectral ordinates remain blocked pending a documented transform or original numeric data; the hybrid-kinetic comparison is outside a direct CGL–LF recalculation. A pre-replacement production lineage has been exercised only as archival pipeline evidence; the merged turbulent-driver replacement changes its forcing/restart-state contract. No current-implementation standard-duration comparison is claimed in this version of the manuscript.

1 Introduction

In a weakly collisional plasma, conservation of the adiabatic invariants can make changes in magnetic-field strength energetically expensive: field amplification and reduction generate a pressure anisotropy that acts back on the turbulent strain. A central question is whether this feedback suppresses the field-strength-changing motions before mirror and firehose regulation dominates the dynamics. MKS24 addressed this question with driven high- β CGL–LF turbulence and found strong active/passive contrasts in instability exposure, transfer, and scale-dependent organization.

The numerical claim is consequential because it combines non-local heat-flux closure, threshold-activated anomalous scattering, and turbulent statistics in an elongated mean-field box. An independent reproduction must therefore match the physical experiment before interpreting a new parameter scan. In particular, MKS24 varies the anomalous limiter rate after a microinstability threshold is crossed; it does not scan a uniform background collision frequency. We retain the latter experiment as later work, rather than confusing it with the published target.

This manuscript integrates the illustrated validation report originally maintained on branch CGL–STS–LF into the current `feature/cgl-landau-fluid` implementation and establishes the reproduction campaign. It documents the AthenaK CGL–LF method, preserves method-level validation figures, states the MKS24 comparison contract, and records the production and reference-data boundaries needed for defensible claims.

The method-level results show that the implemented update follows the intended finite-volume variable strategy and passes checks including a one-dimensional oblique linear initial-value prob-

lem using the Squire et al. Figure 17 background state. The reduced nonlinear figures exercise analysis and plotting paths only: they are not standard-resolution, steady-state, or quantitative MKS24 reproduction results. The current execution and reference-admission status is maintained in `docs/cgl_lf_mks24_reproduction_implementation_plan.md` and in the Sphinx CGL–LF validation guide.

2 MKS24 Target Contract

The directly reproducible target is the CGL–LF simulation component of MKS24. We use a fully periodic domain $[L_x, L_y, L_z] = [1, 1, 2]$, with the background magnetic field along z , standard resolution $192 \times 192 \times 384$, and initial $\beta_0 \in \{1, 10, 100\}$ where required by the mapped published panels. The nominal closure scale is $|k_{\parallel}| = 4\pi/L_{\parallel} = 2\pi$, the background collision frequency is zero, and the standard microinstability treatment is a hard-wall limiter at the mirror and parallel-firehose thresholds, $\beta\Delta = 1$ and $\beta\Delta = -2$. The forced calculations use an Ornstein–Uhlenbeck velocity source with fiducial injection $\dot{E}_K = 0.32\rho_0 v_A^2 L_{\perp}^3$, shell $k/(2\pi/L_{\parallel}) \in [1, 3]$, and power proportional to k^{-2} . Every production calculation must reach at least $t_f = 10L_{\perp}/v_A$; comparisons use declared late-time windows no shorter than $2L_{\perp}/v_A$.

The comparison has two deliberate limitations. First, MKS24 Figure 10 uses a hybrid-kinetic **Pegasus++** calculation and is external context unless that separate dataset or an equivalent kinetic rerun is obtained. Second, dimensionful spectral ordinate comparisons are withheld where the published material does not state the discrete Fourier normalization required for an AthenaK transform. We may therefore establish reproduction of specified CGL–LF products, but we do not claim complete reproduction of every published panel unless these boundaries are closed.

3 Conserved Variables

The hyperbolic CGL system evolves total energy and the conserved anisotropy variable

$$A = \rho \log \left[\frac{p_{\perp}}{p_{\parallel}} \frac{\rho^2}{|B|^3} \right]. \quad (1)$$

The legacy array slot is still named `IMU`, with `IAN` used as an alias in CGL code. The Riemann solve remains in `A`, which avoids the small- $|B|$ pathologies associated with evolving magnetic moment directly.

The LF STS operator temporarily changes only the interpretation of the `IAN` slot:

$$A \rightarrow \mu_B = \frac{p_{\perp}}{|B|}. \quad (2)$$

The standard conserved-variable STS update then advances (E, μ_B) with the LF flux divergence. Immediately after the RKL2 stage, the EOS converts $\mu_B \rightarrow A$, and the normal CGL conserved-to-primitive recovery restores $(p_{\parallel}, p_{\perp})$.

In the current source tree the LF parabolic operator is owned by `src/diffusion/cgl_landau_fluid.*`; the CGL state and constraint logic are implemented in `src/eos/cgl_mhd.cpp` and `src/eos/cgl_physics.hpp`. The quantitative method tests use `src/pgen/tests/cgl_landau_fluid.cpp`, while the reduced MKS24 initialization and history diagnostics use `src/pgen/tests/cgl_lf_paper.cpp`.

4 Landau-Fluid Closure

The implemented closure uses the Snyder–Hammett–Dorland 3+1 heat fluxes,

$$q_{\parallel} = -\chi_{\parallel} \rho \nabla_{\parallel} \left(\frac{p_{\parallel}}{\rho} \right), \quad (3)$$

$$q_{\perp} = -\chi_{\perp} \left[\rho \nabla_{\parallel} \left(\frac{p_{\perp}}{\rho} \right) - p_{\perp} \left(1 - \frac{p_{\perp}}{p_{\parallel}} \right) \frac{\nabla_{\parallel} |B|}{|B|} \right], \quad (4)$$

with the finite-collision Squire et al. coefficients

$$\chi_{\perp} = \frac{2c_{\parallel}^2}{\sqrt{2\pi} c_{\parallel} |k_{\parallel}| + \nu_{\text{eff}}}, \quad \chi_{\parallel} = \frac{8c_{\parallel}^2}{\sqrt{8\pi} c_{\parallel} |k_{\parallel}| + (3\pi - 8)\nu_{\text{eff}}}. \quad (5)$$

Here ν_{eff} is the background collision frequency plus any active mirror/firehose limiter scattering at the face. In the collisionless limit, these reduce to $\chi_{\perp} = \sqrt{2/\pi} c_{\parallel}/|k_{\parallel}|$ and $\chi_{\parallel} = \sqrt{8/\pi} c_{\parallel}/|k_{\parallel}|$. Here `lf_k_parallel` is the closure wavenumber $|k_{\parallel}|$, not a conductivity. In local mode, $c_{\parallel} = \sqrt{p_{\parallel}/\rho}$ is computed from the live face state. In background mode, only this prefactor is frozen to `lf_c_parallel0`; all gradients and thermodynamic variables remain live.

The face fluxes coupled into STS are

$$F_E^i = \hat{b}^i \left(q_{\perp} + \frac{1}{2} q_{\parallel} \right), \quad F_{\mu_B}^i = \hat{b}^i \frac{q_{\perp}}{|B|}. \quad (6)$$

5 Heat-Flux Cap

Following Squire et al. (2023), equations 3.1–3.2, the unlimited heat flux q_L is capped as

$$q = \frac{q_L q_{\text{max}}}{q_{\text{max}} + |q_L|}. \quad (7)$$

The implemented caps are

$$q_{\perp, \text{max}} = \sqrt{\frac{2}{\pi}} c_{\parallel} p_{\perp}, \quad q_{\parallel, \text{max}} = \sqrt{\frac{8}{\pi}} c_{\parallel} p_{\parallel}. \quad (8)$$

The cap is applied after computing the unlimited flux with the collision-aware coefficient, so collisions and limiter scattering suppress the flux before the smooth free-streaming bound is applied.

6 Threshold Policy and Retained Diagnostics

The current implementation separates backward-compatible operator tests from MKS24 production definitions. Existing method checks retain the default oblique-firehose policy. MKS24 decks set `cgl_firehose_threshold = parallel`, selecting $\beta\Delta \leq -2$, together with the mirror threshold $\beta\Delta \geq 1$. Hard-wall MKS24 decks additionally set `limiter_hardwall = true`; newly unstable primitive states are then projected to the selected threshold while preserving $p_{\perp} + p_{\parallel}/2$, rather than relying on finite-rate limiter relaxation to approximate an algebraic wall.

LF-enabled runs append persistent history counters for STS stages, floors, non-finite and non-positive states, ordinary limiter occupancy, emergency hard-bound crossings, heat-flux-cap activity, applied heat-flux contractions, and hard-wall projection applications. When enabled for paper decks, `cgl_lf_record_pressure_work` also retains explicit-RK-applied total and anisotropic CGL pressure-traction work. Strict validation rejects floors, invalid states, or emergency-bound crossings; hard-wall projection activity is an expected model action, not by itself a safety failure.

7 Reproduction Campaign

The production matrix is frozen in `inputs/cgl_lf_paper/mks24_stage_i_manifest.json`. It contains sixteen unique CGL–LF calculations after two required alias decisions: the standard active, Alfvénic $\beta_0 = 10$ output also supplies the nominal heat-flux and $n_\perp = 192$ scale-separation roles, while the standard active, Alfvénic $\beta_0 = 100$ hard-wall output also supplies the hard-wall limiter-scan role. The separately defined active, Alfvénic $\beta_0 = 1$ deck is not executed in this matrix because the displayed MKS24 beta-one compressive-spectrum case is randomly forced.

Cases	Physical variation	Published role
R02--R09	Active/passive; Alfvénic/random; $\beta_0 = 10, 100$	State, PDF, transfer, alignment comparisons
R10--R11	Random compressive variants; $\beta_0 =$ 1, 100	Compressive-spectrum comparison
R12--R13	Stronger/weaker heat-flux scales	Closure-scale sensitivity
R14--R15	$\nu_{\text{lim}} = 20, 200$ at $\beta_0 = 100$	Limiter sensitivity
R16--R17	$n_\perp = 96, 384$ around reused R02	Scale-separation comparison

Table 1: Unique AthenaK CGL–LF production calculations mapped to the MKS24 simulation panels. The manifest records each input file and the analysis roles that reuse an accepted output.

Production is executed on Frontier through `scripts/frontier/cgl_lf_stage_i.py`, separately from the debug-only qualification utility. The production utility admits only mapped input files, archives input, executable, and utility checksums, uses the `batch` partition with its default production `normal` QOS, reserves usage against the Stage I ceiling, and permits only one submitted segment at a time. The first submitted segment, legacy shared-I/O R16/s00.t0.t2 (Frontier job 4674731), terminated cleanly on its wall-clock limit at $t = 1.66204848945 < 2$, with zero strict LF failure counters and 1.253333 accounted node-hours. It is retained as clean partial diagnostic evidence, not an accepted result. Its prolonged $t = 1$ snapshot/checkpoint write motivated a committed rank-local retained output protocol with grouped checksum and analysis support. The fresh rank-local R16/s01.rankio.t0.t2 segment was submitted as job 4675322 from $t = 0$, completed cleanly at $t = 1.87769834013 < 2$, and was recorded as an inspected rank-local clean partial using 1.253056 node-hours. Its complete eight-rank $t = 1$ snapshot/restart boundary incurred no legacy-scale pause. The same-digest continuation R16/s02.rankio.t1p877698.t2 (job 4676557) reached $t = 2.0$, passed inspection with zero strict LF failure counters and complete terminal rank-local products, and used 0.106389 node-hours. It is accepted as the terminal segment of the rank-local $t = 0$ -to-2 prefix, not as a completed $t = 10$ case. A further same-digest continuation through $t = 3.5$, job 4676696, completed in 1.383611 node-hours and passed formal inspection with zero strict LF safety counters, five complete rank-local snapshot groups ending at $t = 3.5$, and a complete terminal rank-local restart group. It extends the accepted prefix, but is not a completed $t = 10$ case. The same-digest continuation through $t = 5.0$, job 4679803, completed in 1.567778 node-hours and passed formal inspection with zero strict LF safety counters, six complete rank-local snapshot groups ending at $t = 5.0$, and a complete terminal rank-local restart group. It extends the accepted prefix but is not a completed $t = 10$ case. The subsequent continuation toward $t = 6.5$, job 4681476, terminated cleanly on its application wall-clock limit at $t = 6.43828031751 < 6.5$. Its formal partial-continuation inspection records zero strict LF safety counters, six complete rank-local snapshot groups, complete terminal rank-local restart products, and 1.753333 accounted node-hours. It is retained as a clean partial, not as an accepted extension of the prefix. A short completion segment from that inspected terminal restart set to

$t = 6.5$, job 4683291, completed in 0.080556 node-hours and passed formal inspection with zero strict LF safety counters and complete terminal rank-local products. It extends the accepted prefix through $t = 6.5$. Continuation R16/s07_rankio_t6p5_t7p5, job 4683918, completed in 1.281944 node-hours, reached $t = 7.5$, and passed formal inspection with zero strict LF safety counters and complete terminal rank-local products. It extends the accepted prefix through $t = 7.5$. Continuation R16/s08_rankio_t7p5_t8p5, job 4686032, completed at exact $t = 8.5$ in 1.282778 node-hours and passed formal inspection with zero strict LF safety counters and complete terminal rank-local products. It was recorded as rejected for current reproduction admission because the merged turbulent-driver replacement changes forcing evolution and restart state/layout. Stage I has consumed 9.962778 node-hours with no active reservation; a replacement-driver production lineage must start at $t = 0$ after qualification and immutable-build archival. The bundle assembler follows only the explicit inspected restart lineage and bounds a sampled-history boundary gap using the retained history cadence and boundary timesteps, rather than mixing in the abandoned shared-output branch. Applied to the ranked $t = 0$ -to-2 prefix, it linked ten distinct rank-local snapshots from only the two rank-local segments. A separately labeled transient-window analysis over $0 \leq t \leq 2$ read all ten snapshots, selected 101 history rows, passed its synthetic diagnostic check, and rendered nine diagnostic figures. This check qualifies the retained-output pipeline only; it is not a late-time MKS24 comparison. After acceptance through $t = 3.5$, a new lineage-checked prefix bundle assembled only the three ranked-output segments and retained fifteen deduplicated rank-local snapshots. This verifies accepted-prefix assembly at the later gate, but it remains outside the declared $8 \leq t \leq 10$ comparison window: a default-window analysis found 176 merged history rows overall but selected zero history rows and zero snapshots for comparison, while passing its synthetic diagnostic check. After acceptance through $t = 5.0$, a further lineage-checked prefix bundle assembled only the four ranked-output segments and retained twenty-one deduplicated rank-local snapshots. A default-window analysis found 251 merged history rows overall but selected zero history rows and zero snapshots in $8 \leq t \leq 10$, while passing its synthetic diagnostic check. After acceptance through $t = 6.5$, the lineage-checked prefix includes the inspected clean-partial continuation and its accepted terminal-gate segment; it retains twenty-eight deduplicated rank-local snapshots. A default-window analysis found 326 merged history rows overall but selected zero history rows and zero snapshots in $8 \leq t \leq 10$, while passing its synthetic diagnostic check. After acceptance through $t = 7.5$, the next lineage-checked prefix retains thirty-one deduplicated rank-local snapshots. Its generic analysis found 376 merged history rows overall but selected zero history rows and zero snapshots in $8 \leq t \leq 10$, while passing its synthetic diagnostic check. A requested Figure 11 comparison is therefore rejected as an empty selected analysis rather than presented as paper evidence.

8 Validation Workflow

Generate the method-validation figures from the repository root with

```
python3 scripts/cgl_lf_workflow.py full --output-dir /path/to/full-bundle
```

or with the compatibility command `scripts/run_cgl_lf_validation.sh`. The driver writes an ignored result bundle containing a manifest, readable summary, logs, histories, machine-readable validation CSV files, and figures. The five method figures included below were regenerated from the retained passing bundle `build-cgl-implementation/cgl_lf_runs/20260524-final-full-v2`.

Each test prints the measured value, reference value or error norm, tolerance, and pass/fail status. The plotted points are read from the same diagnostic CSV files produced by the passing runs.

Generate reduced nonlinear diagnostic products with

```
python3 scripts/cgl_lf_workflow.py paper-convergence
python3 scripts/cgl_lf_workflow.py paper-analyze \
--output-dir /path/to/paper-convergence-bundle
```

The representative nonlinear figures below were regenerated with the current analyzer from the retained passing 20260525-paper-convergence-pressure-work-v1 bundle. Its cases run only through $t = 0.02$ on reduced grids and therefore test product plumbing, not statistically saturated turbulence or paper agreement.

9 Validation Problems and Tolerances

10 Quantitative Results

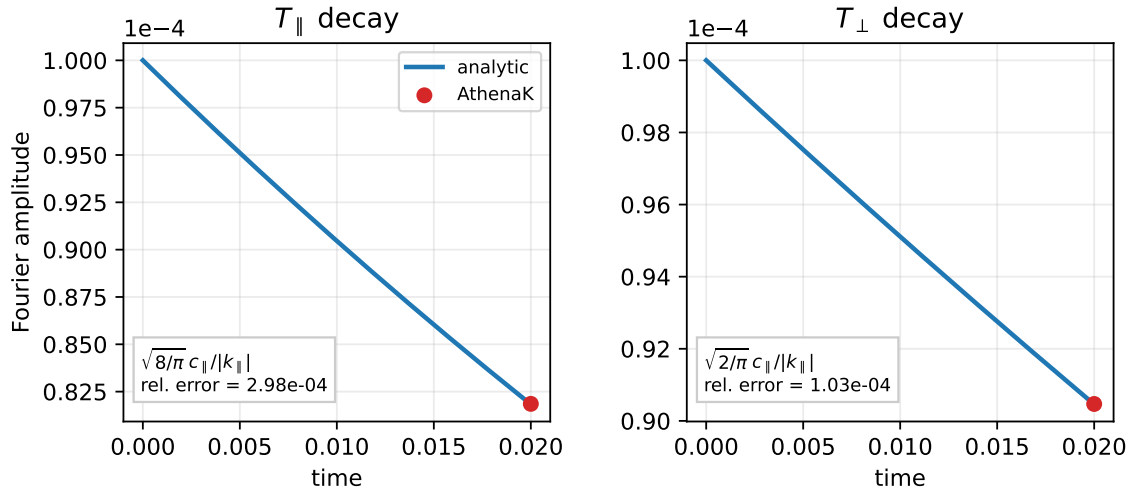


Figure 1: Constant-coefficient decay tests. The $T_{\parallel}$ case uses $\chi_{\parallel} = \sqrt{8/\pi} c_{\parallel}/|k_{\parallel}|$; the $T_{\perp}$ case uses $\chi_{\perp} = \sqrt{2/\pi} c_{\parallel}/|k_{\parallel}|$. The red point is the final AthenaK Fourier amplitude and the blue curve is the analytic constant-coefficient decay.

Input	Purpose	Assertion
cgl_transform_check	$A \leftrightarrow \mu_B$ and primitive recovery	pass
cgl_lf_quant_parallel	sinusoidal $T_{\parallel}$ decay	final Fourier amplitude within 5% of analytic decay
cgl_lf_quant_parallel_collisional	sinusoidal $T_{\parallel}$ decay with $\nu_c > 0$	final Fourier amplitude within 7% of finite-collision analytic decay
cgl_lf_quant_perp	sinusoidal $T_{\perp}$ decay	final Fourier amplitude within 5% of analytic decay
cgl_lf_quant_perp_collisional	sinusoidal $T_{\perp}$ decay with $\nu_c > 0$	final Fourier amplitude within 7% of finite-collision analytic decay
cgl_lf_quant_grad_b	isolated $\nabla_{\parallel} B $ term	RMS relative error below 10%, positive response correlation
cgl_lf_flux_limiter	equation 3.2 heat-flux cap	$ q \leq q_{\max}$, sign preserved, update within 20%
cgl_lf_limiter_heat_flux_suppression	limiter-collision suppression of heat fluxes	active limiter scattering reduces the unlimited LF flux by more than 20x
cgl_lf_limiter_mirror	mirror-biased admissibility stress	intermediate STS states finite, positive, and backup bounded
cgl_lf_limiter_firehose	firehose-biased admissibility stress	intermediate STS states finite, positive, and backup bounded
cgl_lf_field_wave	field-aligned LF wave reference	complex Fourier mode within 25% of offline reference
cgl_lf_paper_oblique_wave	Figure 17 background, LF enabled	guard tolerance 40%; observed errors $< 3 \times 10^{-6}$
cgl_pure_paper_oblique_wave	Figure 17 background, no LF	guard tolerance 40%; observed errors $< 3 \times 10^{-6}$
cgl_pure_paper_eigen_*	exact pure-CGL Alfvén/slow/fast modes	all tracked Fourier components within 7.5%
cgl_lf_paper_eigen_*	exact CGL-LF Alfvén/slow/fast modes	all tracked Fourier components within 7.5%

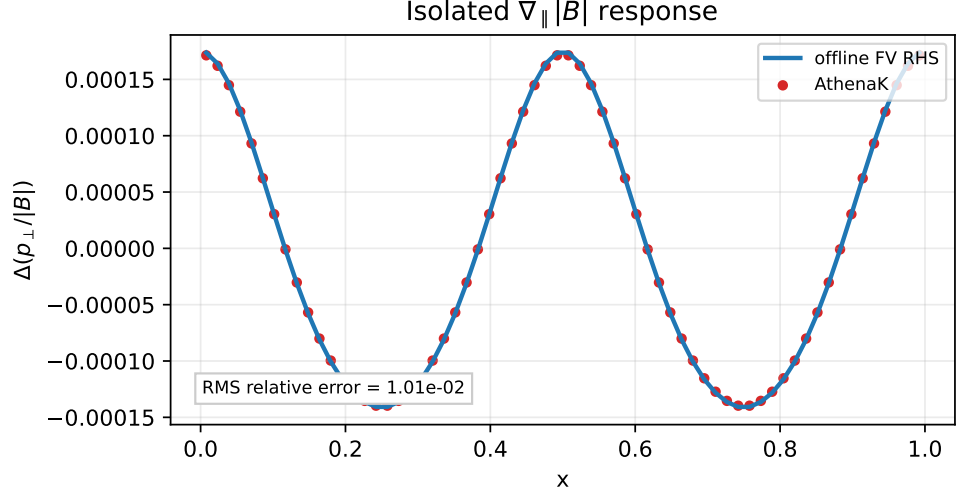


Figure 2: Isolated $\nabla_{||}|B|$ term in the perpendicular heat flux. The initial thermal gradients are zero; the plotted response is the short-time change in $p_{\perp}/|B|$ compared with an offline finite-volume reconstruction of the initial LF RHS.

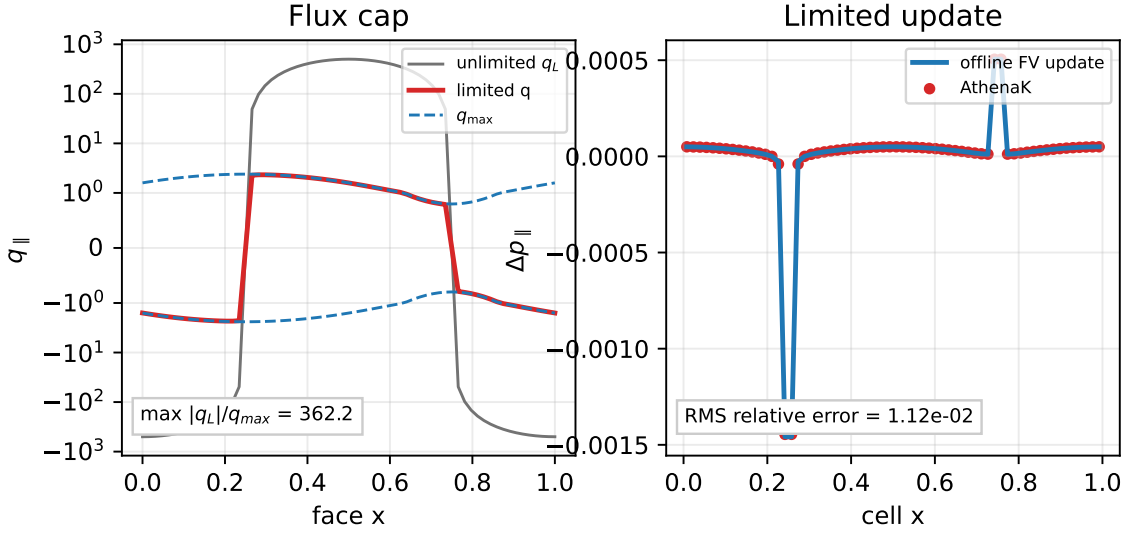


Figure 3: Heat-flux limiter test. The left panel shows the unlimited $q_{||,L}$, the capped $q_{||}$, and $\pm q_{||,\max}$. The right panel checks that the pressure update produced by the limited flux matches the offline finite-volume update.

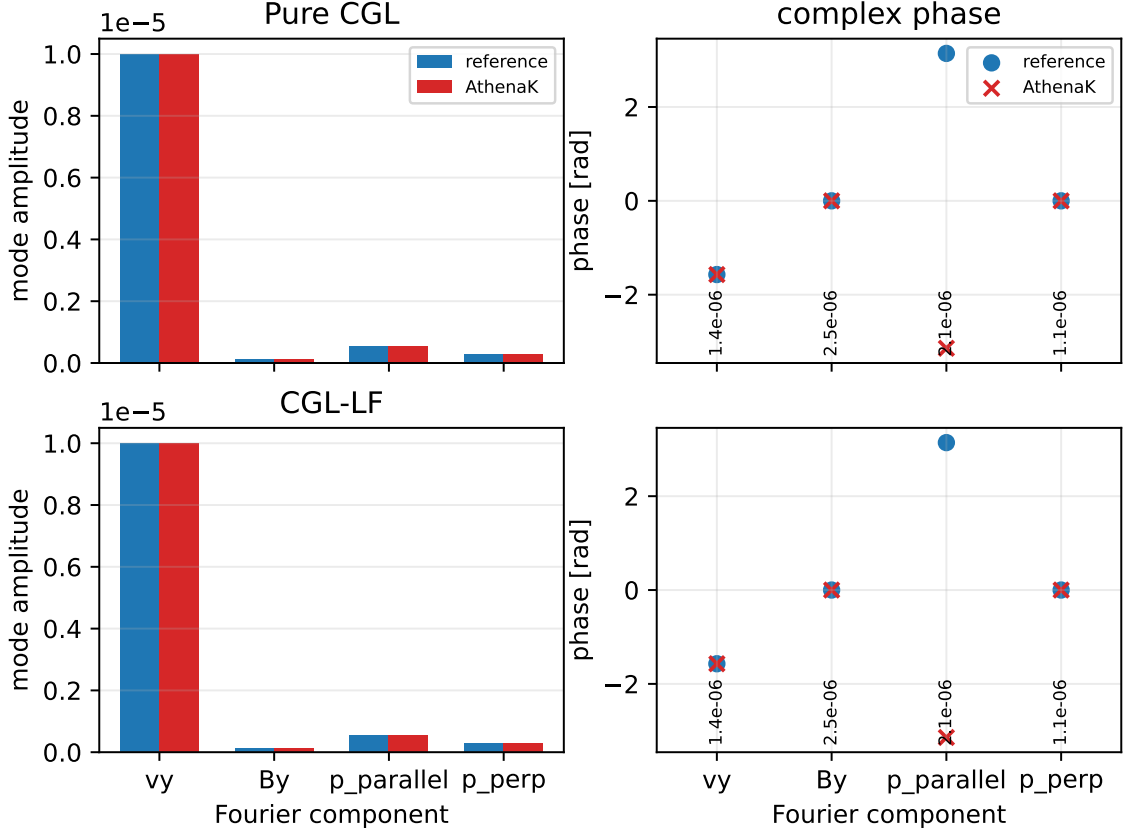


Figure 4: Figure 17-style oblique linear initial-value check. The plotted quantities are the final complex Fourier amplitudes of u_y , B_y , $p_{\parallel}$, and $p_{\perp}$ for the pure-CGL and CGL-LF inputs, compared with the offline linearized initial-value reference used by the quantitative pgen. The vertical phase-panel labels are per-variable relative errors.

11 Figure 17-Style Linear Initial-Value Check

The paper-style inputs use the Figure 17 background

$$\rho_0 = 1, \quad p_{\parallel 0} = p_{\perp 0} = 5, \quad u_0 = 0, \quad B_0 = (1, \sqrt{2}, 0.5), \quad k = 2\pi\hat{x}, \quad kL = |k|. \quad (9)$$

The test initializes a small transverse velocity perturbation, not a CGL eigenmode, and compares the final complex Fourier amplitudes of u_y , B_y , $p_{\parallel}$, and $p_{\perp}$ against an offline linearized CGL–LF initial-value reference integrated with the same background state and closure coefficients. Two inputs are provided: one with the LF heat flux enabled and one with pure CGL. This checks the same method class as the Appendix A.2/Figure 17 linear-wave convergence problem.

12 Figure 17-Style Eigenmode Checks

The validation suite now also includes exact eigenmode initializations for the Alfvén, slow-like, and fast-like positive-frequency branches of the same linearized Figure 17 system. The script `scripts/generate_cgl_lf_eigenmode_inputs.py` constructs the linear CGL or CGL–LF matrix for the AthenaK variables $(\rho, u_x, u_y, u_z, B_y, B_z, p_{\parallel}, p_{\perp})$, diagonalizes it, normalizes the selected eigenvectors, and writes their complex amplitudes and eigenvalues into AthenaK input files. The pgen initializes $\text{Re}[\delta U_k \exp(ikx)]$ and compares the final Fourier mode with $\delta U_k \exp(\lambda t)$, where λ is the supplied eigenvalue. These inputs are the direct replacement for using a single transverse-velocity seed as a proxy for the Squire/Jono linear-wave comparisons.

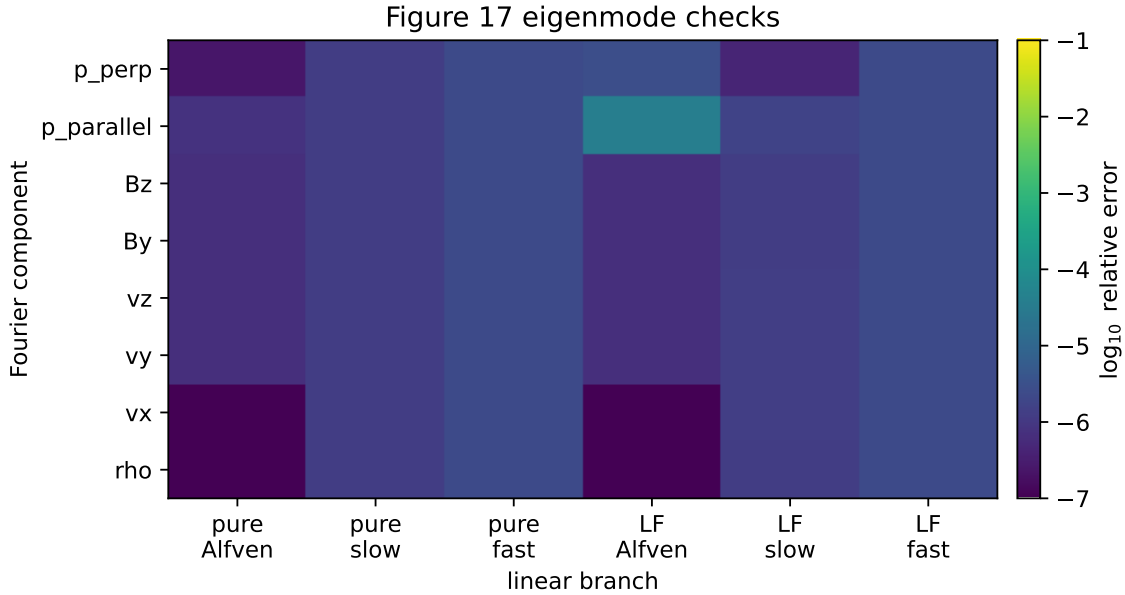


Figure 5: Exact eigenmode checks on the Figure 17 background. Each column is a pure-CGL or CGL–LF Alfvén, slow-like, or fast-like positive-frequency branch. The color scale shows the complex Fourier relative error in each primitive or magnetic component after a short AthenaK evolution compared with the supplied linear eigenmode prediction $\delta U_k \exp(\lambda t)$. The white numbers give the maximum tracked-component error in each branch.

13 Reduced MKS24-Oriented Analysis Products

The `paper-convergence` workflow supplies short, reduced forced turbulence bundles used to validate nonlinear diagnostics and rendering. The figures in this section use the current analyzer on retained snapshots from `20260525-paper-convergence-pressure-work-v1`. The bundle contains three resolution-path cases and heat-flux/policy variants, all with clean LF safety diagnostics. Because these cases terminate at $t = 0.02$, their curves must not be interpreted as steady turbulent spectra or MKS24 comparisons.

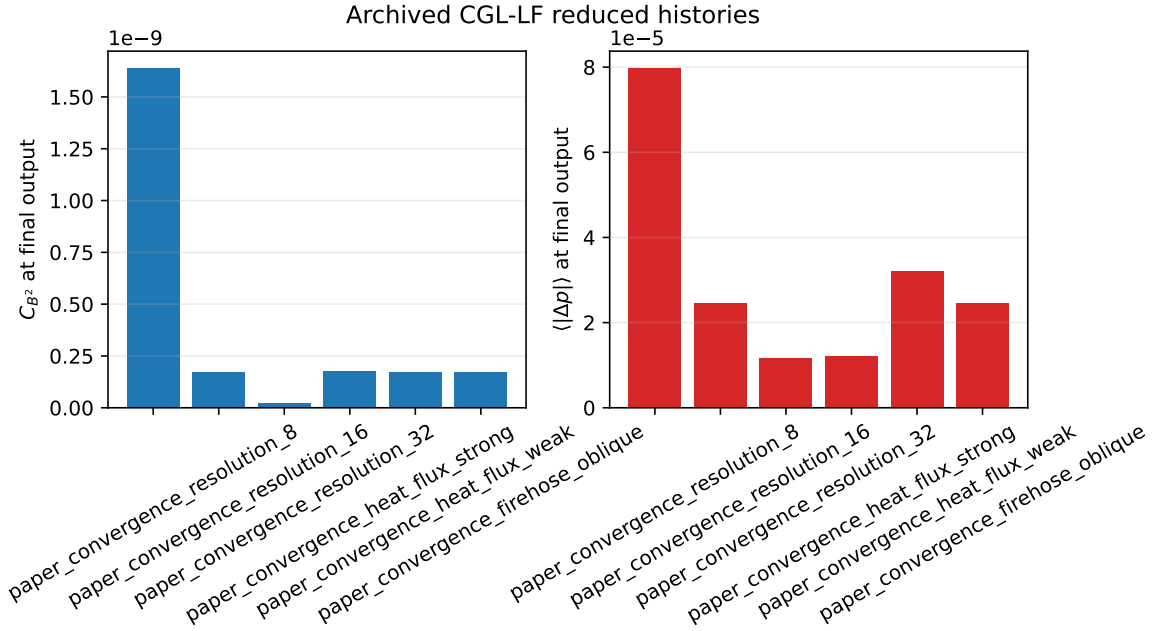


Figure 6: Reduced history-product rendering for the nonlinear analysis path. The plotted global summaries verify ingestion of archived history data across the reduced case matrix; they do not establish converged physical trends.

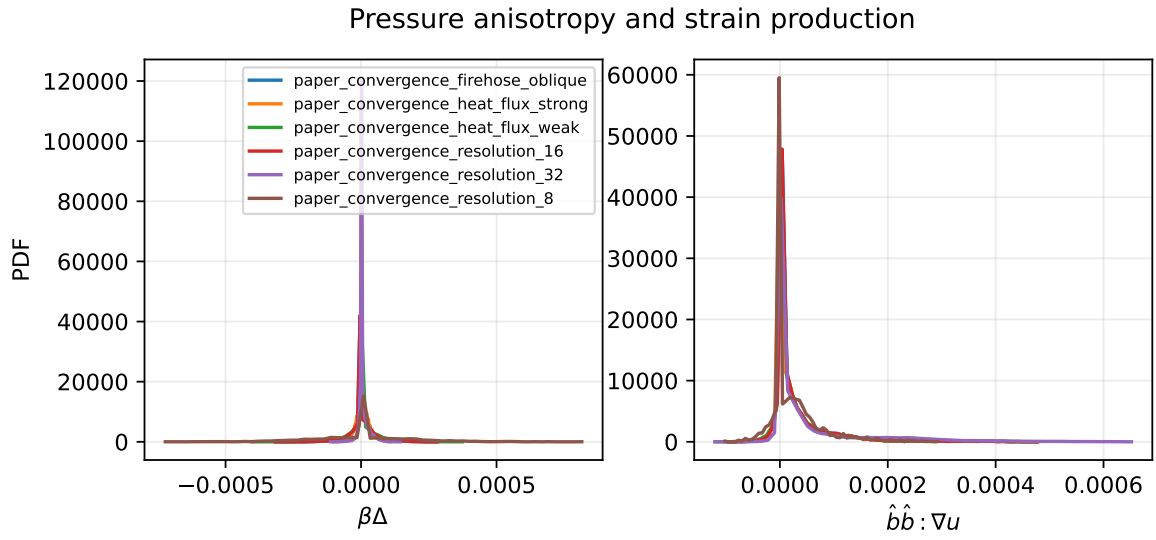


Figure 7: Reduced pressure-anisotropy and local-strain PDF rendering. These products expose $\beta\Delta$ and $\hat{\mathbf{b}}\hat{\mathbf{b}} : \nabla \mathbf{u}$ for later panel comparisons once authorized statistically useful runs exist.

Pressure-density joint PDF: paper_convergence_resolution_32

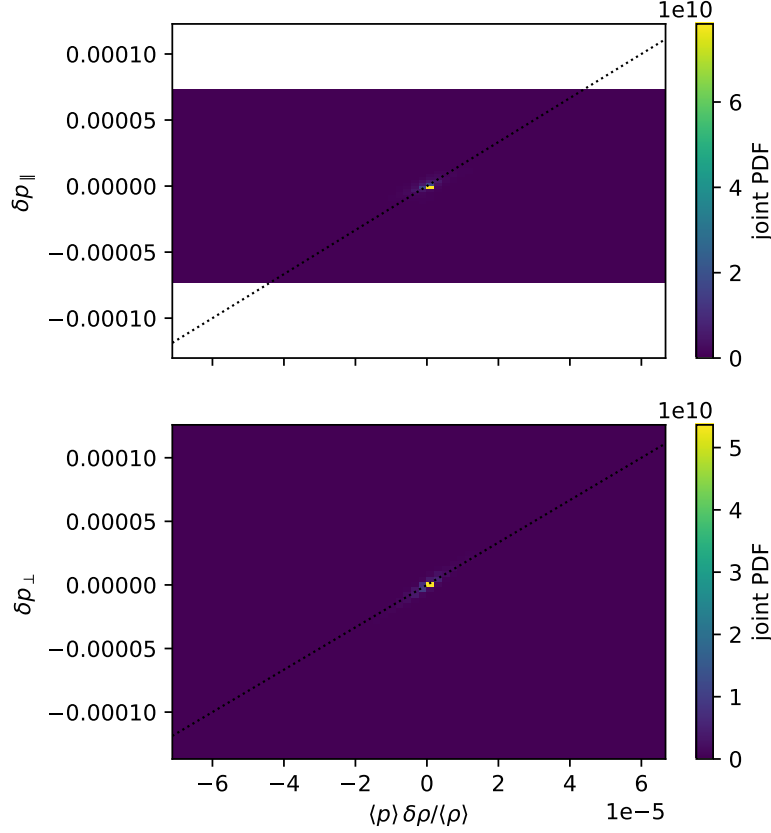


Figure 8: Representative Figure 2(a)-coordinate joint-PDF diagnostic from the $32 \times 32 \times 64$ reduced case. The analyzer now emits $\langle p \rangle \delta \rho / \langle \rho \rangle$ versus $\delta p_{||}$ and $\delta p_{\perp}$. The admitted digitized MKS24 surface route is separate from this reduced simulation output and requires a matching standard case.

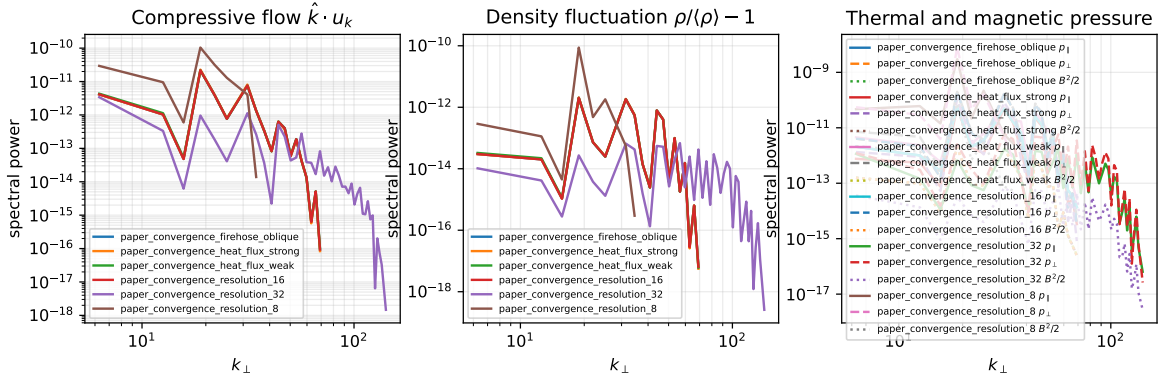


Figure 9: Current reduced rendering of the Figure 3/4(b)/6(a)-oriented spectral fields: compressive velocity projection, normalized density fluctuation, and thermal/magnetic-pressure products. Their plotted MKS24 absolute ordinates remain excluded from reference comparison until an external discrete Fourier-normalization mapping or original numeric data are available.

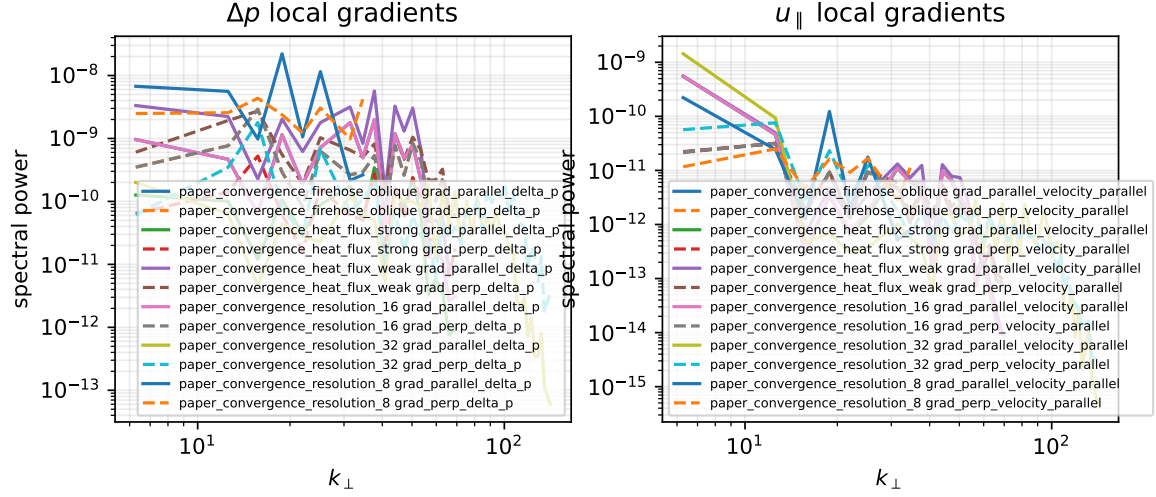


Figure 10: Projected pressure-anisotropy and velocity-gradient spectrum products from the reduced bundle. These are diagnostic-product checks and not admitted absolute-ordinate paper comparisons.

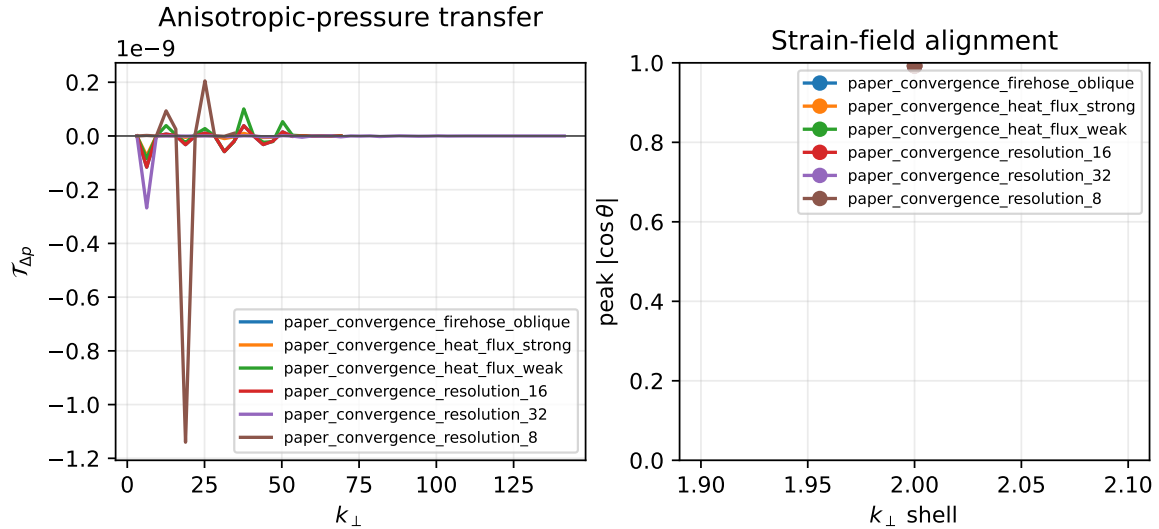


Figure 11: Pressure-stress transfer and alignment-product rendering. The analyzer supports the dimensionless MKS24-normalized transfer comparison and selected-shell or peak-alignment comparisons when supplied with provenance-qualified references and matching executed cases.

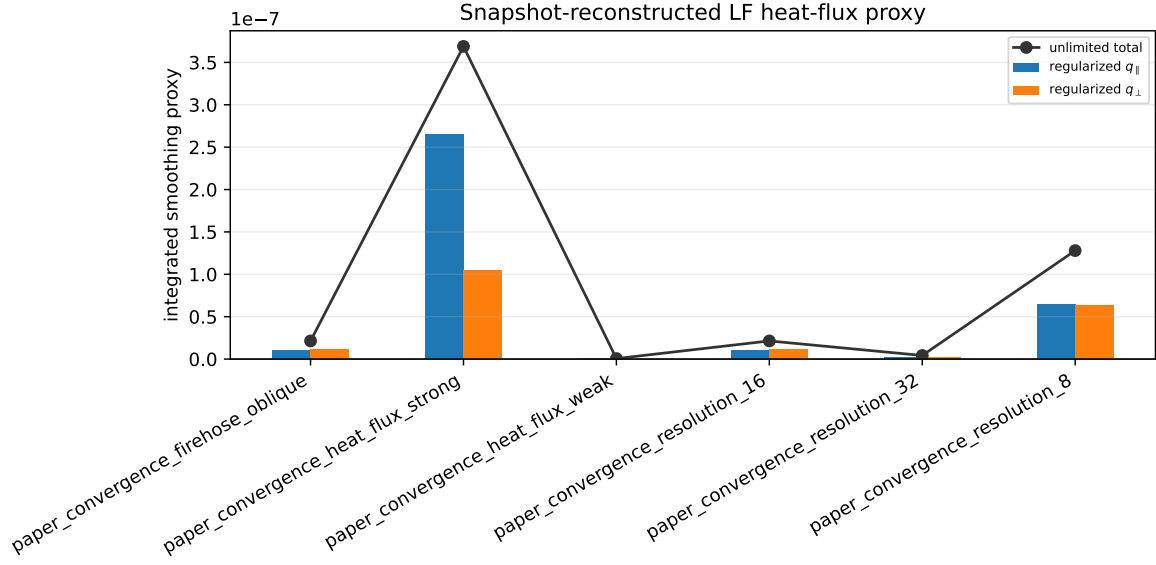


Figure 12: Snapshot-reconstructed heat-flux smoothing proxy for reduced cases. It is useful for interpretation of retained snapshots, but is not the applied finite-volume heat-flux contraction or a total energy closure.

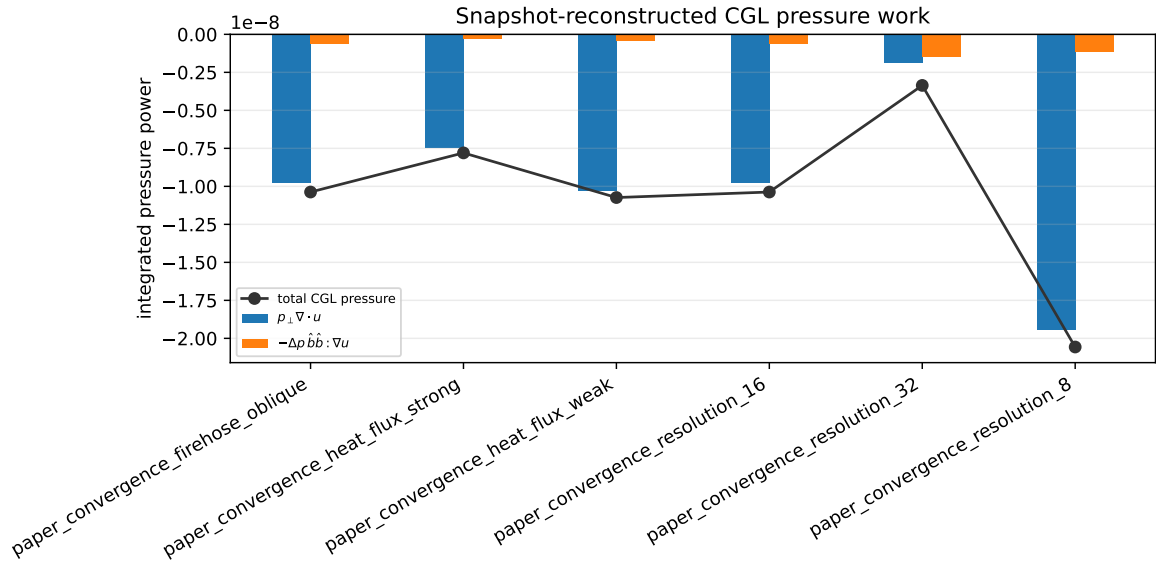


Figure 13: Snapshot-derived CGL pressure-work decomposition, including the anisotropic stress contribution checked against direct pressure transfer. Paper decks also retain applied pressure-traction ledger histories; neither the reduced snapshot plot nor an unevaluated standard deck establishes a production result.

14 MKS24 Reproduction Boundary

The current branch defines guarded standard, limiter-frequency, heat-flux, compressive, and scale-separation workflows and supports checksum/uncertainty-qualified external reference comparisons. The admitted published reference products presently include Figure 2(a) transformed sampled surfaces, Figure 2(b) histories, normalized Figure 4(a) PDFs, normalized Figure 5(b) eddy scales, dimensionless Figure 7 transfer ratios, Figure 8 selected-shell alignment PDFs, Figure 9 and Figure 11/12 alignment curves, and Figure 13(b),(d) dimensionless products. Because each admitted panel product retains its own provenance manifest, an accepted campaign bundle is analyzed by repeating `paper-analyze --reference-curves` for the admitted manifests; the analyzer records every manifest checksum and provenance block and rejects duplicate product identifiers across manifests. For incremental review of a single accepted case, `--allow-partial-reference-cases` evaluates only products whose cases are present in the bundle and records the omitted products; it does not authorize a missing-case panel claim. Snapshot-window selection reads the binary output time from each header before loading fields, so retained snapshots outside the declared late-time comparison window do not dominate analysis cost.

Remaining absolute spectral or strain ordinate panels are intentionally not admitted: the paper and cited donor source define shell spectra and plotted units but do not state the discrete Fourier normalization needed to map absolute digitized ordinates into AthenaK products. Those panels require author/archive numeric data, donor diagnostic code, or an explicit normalization statement. One mapped production segment has been retained only as clean partial I/O-diagnostic evidence; the ranked-output replacement is accepted through its $t = 7.5$ prefix. Its first continuation beyond $t = 5.0$ is retained as a clean partial through $t = 6.43828031751$, and its inspected short terminal-gate completion establishes the accepted $t = 6.5$ prefix. The accepted continuation to $t = 7.5$ remains outside the comparison window. A pre-replacement continuation reached $t = 8.5$ cleanly but is rejected for current reproduction admission after the turbulent-driver forcing/restart-state contract changed. Its transient-prefix bundle has verified the rank-local assembly and diagnostic path, but the production comparison window is $8 \leq t \leq 10$. Quantitative MKS24 simulation-to-reference comparisons require completed accepted bundles; none is claimed by this manuscript version.

15 Comparison Gate Matrix

Table 2 records the evidentiary boundary for every published-panel family targeted by Stage I. An “admitted” comparison has a declared dimensionless transform and an archived digitized or analytic reference; it still requires a matched, accepted AthenaK output bundle. “Blocked” means that a necessary reference normalization or external simulation product is not available in the present reproduction archive.

16 Discussion

The reproduction is designed to distinguish a physical active-pressure response from two easier but weaker demonstrations: a correct implementation of individual LF terms and a visually plausible short turbulence transient. The validation results establish the first requirement at the operator and linear-wave levels. The guarded Stage I matrix and the comparison gates make the second requirement testable, but it cannot be resolved by the reduced diagnostic bundles shown above.

MKS24 panel	AthenaK cases	Quantitative gate	Status
Figure 1	R02, R06	Common-time active/passive spatial comparison; qualitative captioned interpretation only.	Awaiting runs
Figure 2(a–b)	R02--R09	Admitted sampled-surface and unstable-volume reference products.	Awaiting runs
Figure 3	R03--R05, R09--R11	Absolute compressive spectral ordinate normalization not established.	Blocked/reference
Figure 4	R02--R09	Density PDF admitted; density spectral ordinate mapping blocked.	Partial gate
Figures 5–6	R02, R04	Eddy-scale product admitted where normalized; absolute spectra blocked.	Partial gate
Figures 7–9	R02--R09	Dimensionless transfer and alignment reference products admitted.	Awaiting runs
Figure 10	none	External Pegasus++ hybrid-kinetic comparator.	External data
Figure 11	R16, R02, R17	Dimensionless scale-separation alignment product admitted.	R16 active
Figure 12	R12, R02, R13	Dimensionless heat-flux-scale alignment product admitted.	Awaiting runs
Figure 13	R03, R07, R14--R15	Admitted dimensionless limiter-sensitivity products only.	Awaiting runs

Table 2: Stage I panel-family comparison gates. “Awaiting runs” does not denote agreement or disagreement; it denotes absence of accepted standard-duration AthenaK bundles.

Evidence for the MKS24 interpretation will require late-time active/passive contrasts in the admitted instability, transfer, alignment, and sensitivity products from accepted production data.

Two limitations are structural rather than statistical. The spectral ordinate panels cannot be used for absolute residual tests until the published normalization is mapped to the AthenaK transform or original numeric products are recovered. Similarly, no CGL–LF rerun is an independent reproduction of the hybrid-kinetic Figure 10 comparator. These limitations narrow the claims supported by Stage I but do not prevent quantitative evaluation of the admitted CGL–LF diagnostics.

The later weak-guide-field and background-collisionality study is therefore not part of the present results. In particular, varying uniform background collisionality would answer a different question from changing the threshold-activated limiter rate in MKS24. Treating those scans as interchangeable before the reproduction baseline is accepted would obscure both the numerical and physical comparison.

17 Conclusions

AthenaK now has a validated CGL–LF implementation path, retained nonlinear diagnostics, source-mapped MKS24 case definitions, and a checksum- and allocation-recorded Frontier production protocol. The production protocol began with the required low-resolution scale-separation case; its first shared-I/O segment is a clean partial diagnostic record, and its ranked-output replacement passed the inspected $t = 7.5$ prefix gate. Its clean $t = 8.5$ continuation is retained only as pre-replacement archival evidence after the turbulent-driver contract changed. No completed case is yet counted as an accepted comparison result. Thus the present manuscript establishes the method, target contract, and acceptance procedure, but does not yet claim reproduction of the MKS24 turbulent conclusions.

Completion of the reproduction claim requires all sixteen mapped cases to reach $t_f \geq 10L_\perp/v_A$, pass strict safety and provenance checks, and be evaluated against each admitted comparison gate. Results of the later extension campaign will be reported only after that baseline is complete and a separate execution plan is approved.

18 Data and Reproducibility

The frozen Stage I matrix is stored in `inputs/cgl_lf_paper/mks24_stage_i_manifest.json`; preparation, submission accounting, and terminal-time acceptance inspection are performed by `scripts/frontier/cgl_lf_stage_i.py`. The first production executable was built from revision `e129a7ff8be60eb177c8268da0a73e7700d8b543`, with archived executable SHA-256 digest `de99066f3546976093f6ff656660952032e30f31c939736c0212629cfb015157`. For each segment, the production record retains the submitted input, executable and input provenance, Slurm allocation record, histories, snapshots, restarts, and an inspection record that tests target time and strict LF safety counters before acceptance. Production products are retained beneath the project `runs/mks24-stage-i` archive and will be linked to quantitative panel products only after their corresponding gates pass. Following the measured shared-I/O delay in job `4674731`, commit `1f52784d291718fff3bf1dc984a682ada52780ab` stores production snapshots and restarts as rank-local product sets; inspection retains every rank-file digest and analysis reconstructs the combined snapshot fields. Commits `476f9dbd`, `9c5c1b40`, and `bad8ba05` make analysis bundles follow inspected restart ancestry, permit the expected sampled output boundary without requiring duplicated history rows, and reject larger gaps inconsistent with the recorded history cadence. Commit `449e297b` records absent comparison cases explicitly during scoped single-case analysis and selects snapshot windows from lightweight binary-header reads before reconstructing retained fields.

19 Build and Runtime Status

The imported method figures come from a passing local `full` workflow bundle. The reduced nonlinear figures come from a passing `paper-convergence` bundle reanalyzed with the current plotting code. Warning-strict Sphinx documentation and this L^AT_EX build are checked when the report is updated. The ranked-output low-resolution case passed its pre-replacement inspected $t = 7.5$ prefix gate after a clean partial at $t = 6.43828031751 < 6.5$ and its formally accepted terminal-gate continuation. Its subsequent $t = 8.5$ segment completed cleanly but is rejected for current reproduction admission because the replacement driver changes forcing/restart state; a fresh qualified lineage is required. Nevertheless, production/statistical and paper-reference comparison gates remain open until accepted $t \geq 10$ output bundles are analyzed.

References

- [1] J. Squire et al., *Pressure anisotropy and viscous heating in weakly collisional plasma turbulence*, arXiv:2303.00468, 2023.
- [2] S. Majeski, M. W. Kunz, and J. Squire, *Self-organization in collisionless, high- β turbulence*, arXiv:2405.02418, 2024.